

LOKESH L

Artificial Intelligence / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 2+ years of hands-on experience designing and deploying production-grade machine learning systems. Specialized in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), and multi-agent AI architectures on Microsoft Azure. Proven ability to build scalable end-to-end ML pipelines using vector databases, embedding models, and LLM-based inference. Experienced in Text-to-SQL agent pipelines, PDF data extraction, and custom monitoring solutions. Passionate about building efficient, real-world AI solutions that drive measurable business impact. Skilled in model deployment, API development, and scalable AI system design.

TECHNICAL SKILLS

AI / ML: Machine Learning, Deep Learning, LLMs, NLP, RAG, Multi-Agent AI Systems, Agent-Based Systems, Prompt Engineering, Text-to-SQL, Computer Vision

Frameworks & Libraries: TensorFlow, Scikit-learn, LangChain, Hugging Face Transformers

Cloud & Databases: Microsoft Azure, Azure Cosmos DB, Vector Databases (FAISS, Chroma, Pinecone), SQL

Programming Languages: Python, R, Java, C/C++

MLOps & Tools: Model Deployment, Docker, CI/CD, REST API Development, Embedding Models, Pipeline Design, JSON Logging & Monitoring, Feature Engineering, Model Evaluation, Git

WORK EXPERIENCE

Tata Consultancy Services, Bengaluru, India

July 2024 – Present

AI Engineer

- **Technologies Used:** Python, Azure, LangChain, FAISS, Pinecone, SQL, REST APIs
- Designed and deployed end-to-end Retrieval-Augmented Generation (RAG) systems on Microsoft Azure using vector databases and LLM inference, improving retrieval accuracy by ~25%.
- Built robust data pipelines to extract and preprocess structured and unstructured data from large-scale PDF documents for ML and NLP workflows.
- Engineered multi-agent AI systems to automate enterprise workflows, reducing manual effort by ~30% and improving process efficiency.
- Developed scalable ML pipelines with embedding models and vector databases, enabling efficient semantic search and document understanding.
- Implemented JSON-based logging and real-time monitoring using Azure Cosmos DB to track model performance and system health.
- Applied NLP techniques (NER, information extraction, classification) for large-scale unstructured data processing.
- Built Text-to-SQL pipelines enabling natural language querying for non-technical stakeholders.
- Optimized model and performance through efficient data processing and retrieval strategies, improving system scalability, and response time.
- Collaborated with cross-functional teams to deliver production-ready AI/ML solutions aligned with business goals.

OEConnection, Chennai, India July 2023 – October 2023

AI/ML Developer Intern

- **Technologies Used:** Python, TensorFlow, Scikit-learn, R, Java
- Contributed to the design and development of AI/ML classification and prediction models aligned with business and project requirements.
- Performed data preprocessing, cleaning, and validation, improving data quality and increasing model performance by ~15%.
- Built and evaluated machine learning models using TensorFlow and Scikit-learn, applying best practices in training, hyperparameter tuning, and performance optimization.
- Developed AI/ML solutions in Python, R, and Java, demonstrating cross-language proficiency for diverse use cases.

PROJECTS

Image Caption Generator April 2023

- Built an end-to-end image captioning system combining Computer Vision and NLP using CNN-based feature extraction and sequence-to-sequence models.
- Collected and processed a custom image-text dataset from Twitter, including data cleaning, tokenization, and preprocessing for training and evaluation.
- Achieved 73% caption-to-image matching accuracy using optimized transfer learning and feature extraction within a 4–5 month timeline.

EDUCATION & CERTIFICATIONS

M.Tech (Integrated), Computer Science Engineering — Majors: Business Analytics

Vellore Institute of Technology, Chennai

Foundation in Data Science — Majors: Data Science

Indian Institute of Technology, Madras

LEADERSHIP & ACHIEVEMENTS

Head of Research and Development — Tech Researchers Club, VIT Chennai Sept 2022 – June 2023

- Led a cross-functional R&D team to execute AI/ML innovation projects, aligning research initiatives with organizational goals.
- Mentored team members and fostered a collaborative, problem-solving culture, enhancing overall technical capabilities.